Mtherios Factions

Visual Data Model

This document maps the current codebase as it exists now: a client-first Svelte/Dexie text-adventure engine with an emerging Postgres-backed canon, memory, and sync layer.

Core read

The browser still owns live narration. Dexie holds the local play state. The backend stores canonical-style entities, events, patches, memory nodes, sync operations, import/export, and retrieval packets.

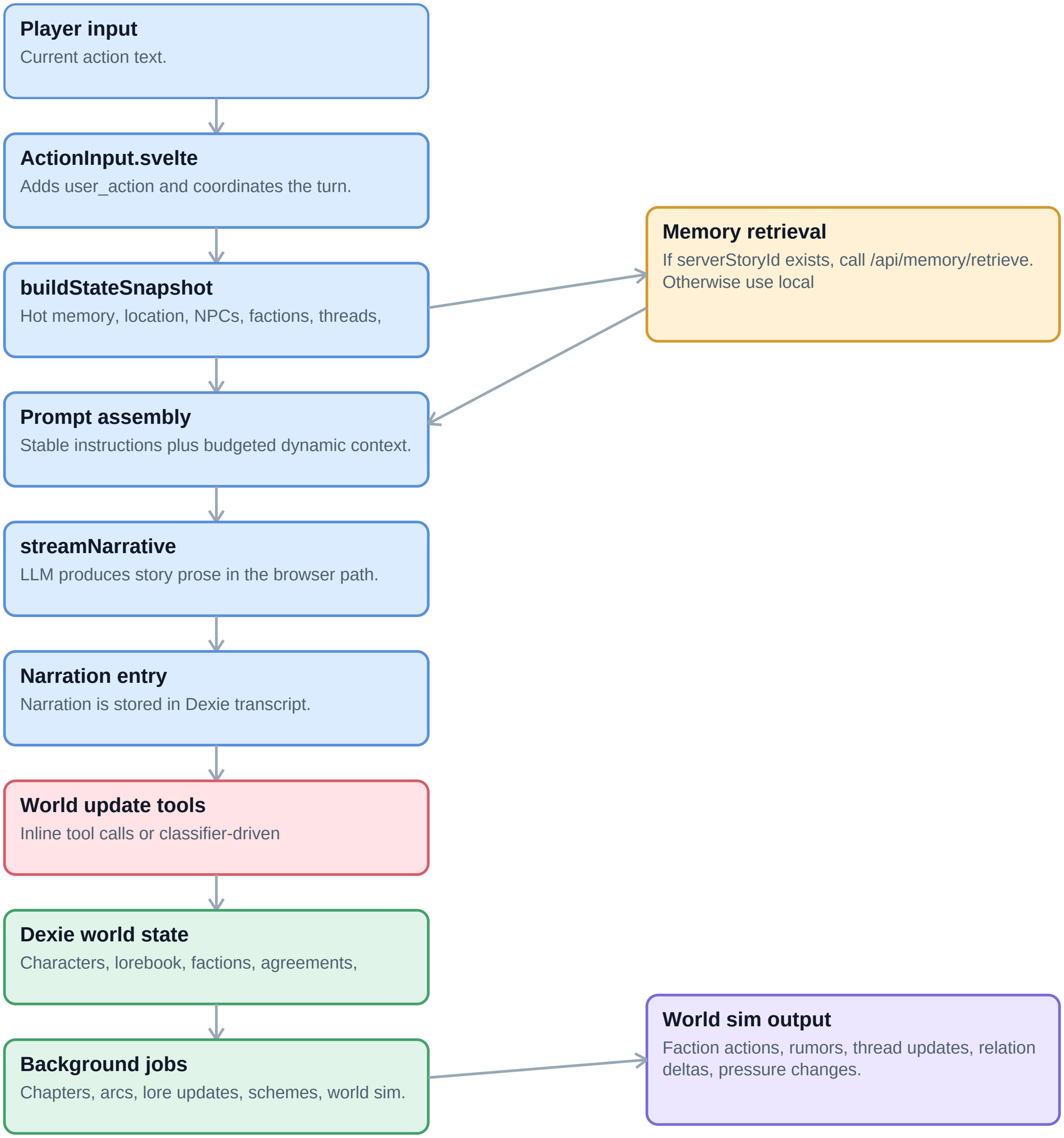
Legend

Blue: prompt/runtime **Green: durable state** **Amber: memory/summary** **Rose: update/sync**

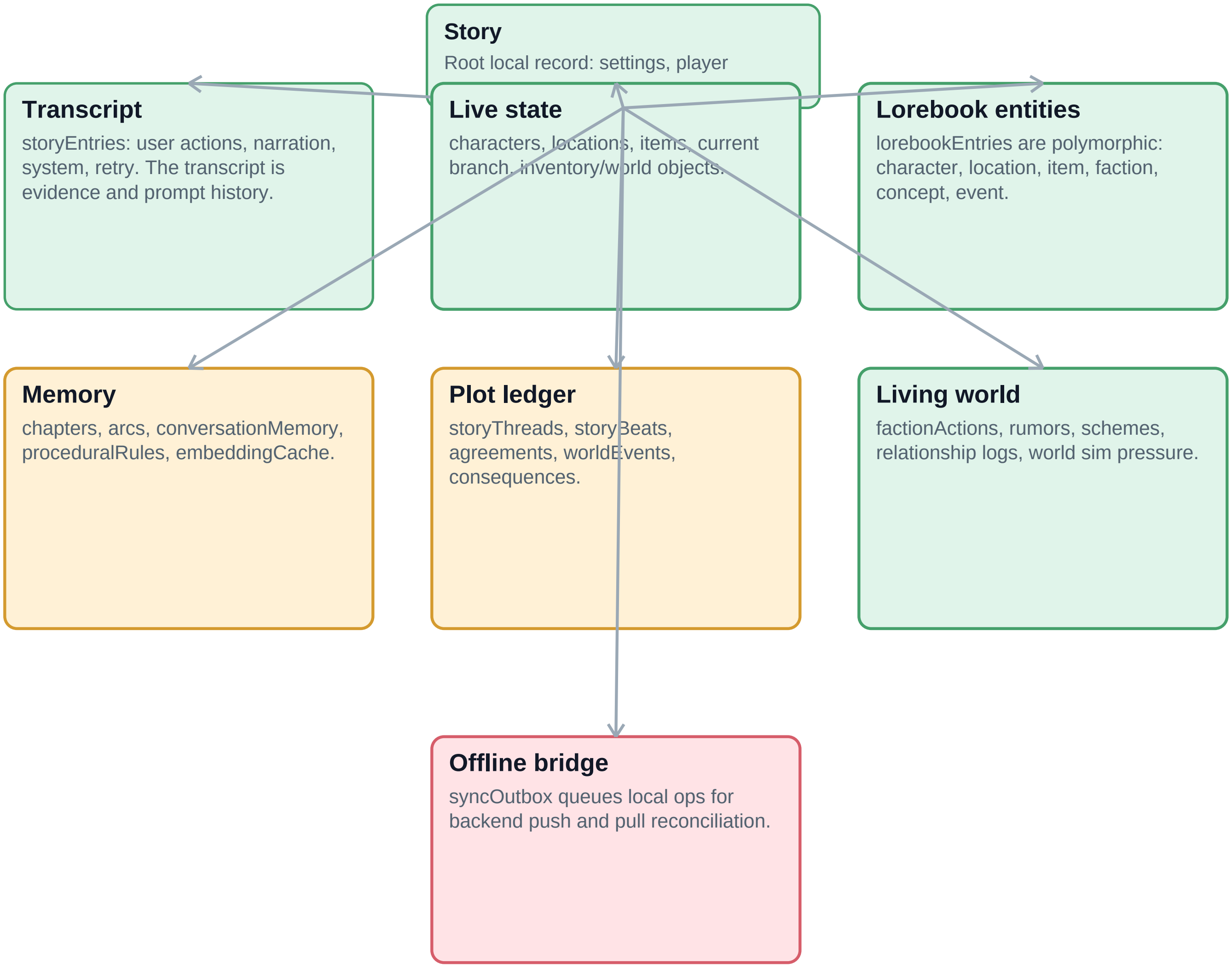
Important files

- src/lib/stores/story.svelte.ts - hot state snapshots, prompt sections, dynamic context budget.
- src/lib/components/story/ActionInput.svelte - current browser-owned turn pipeline and narration streaming.
- src/lib/services/database.ts - Dexie schema and local persistence tables.
- src/lib/server/db/schema.ts - backend Postgres canon, source-linked events, patches, and memory nodes.
- src/lib/server/memory/retrieval.ts - backend hybrid retrieval candidate generation and packet building.

Turn Runtime Flow



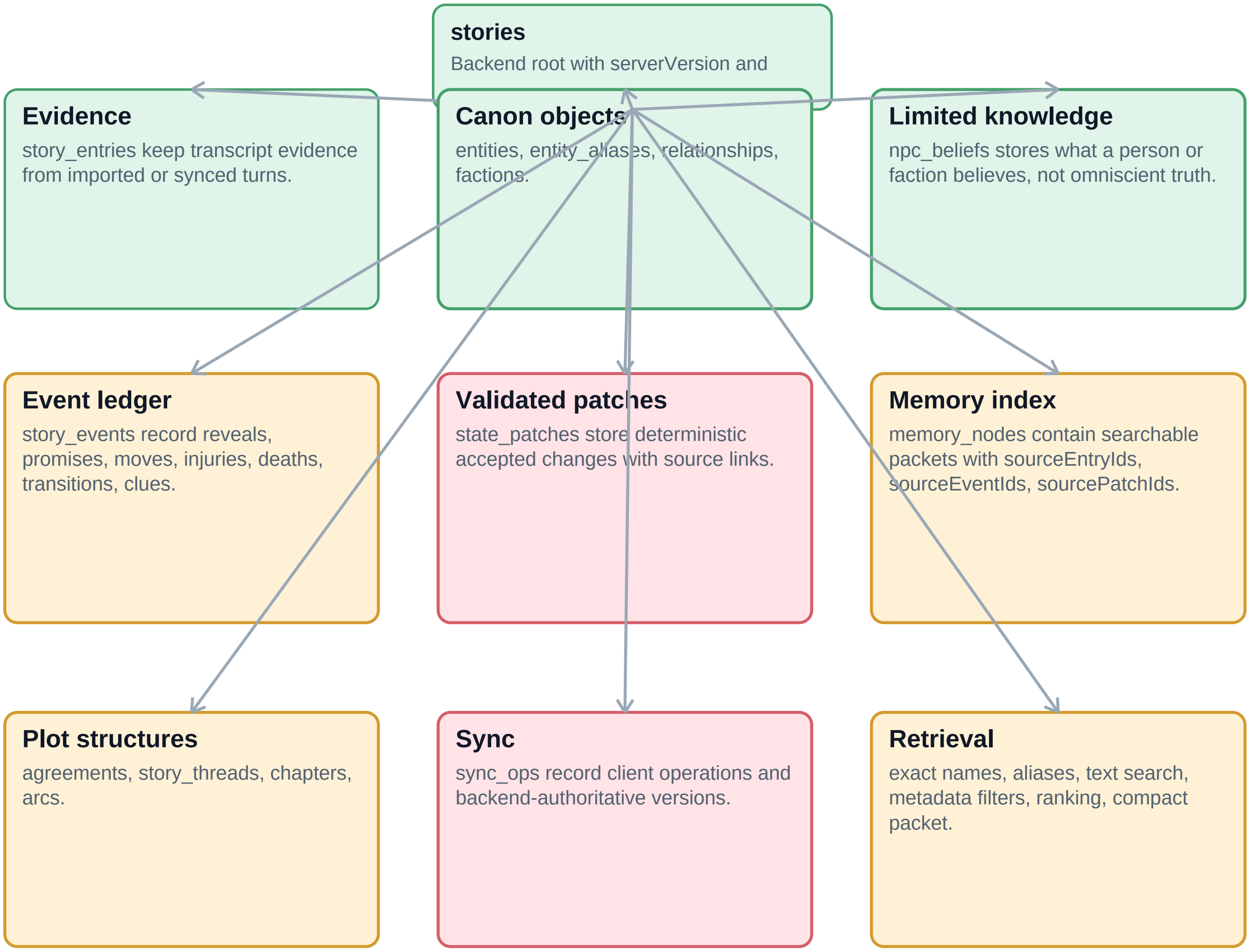
Local Browser Data Model



Local model note

This side is dense because it grew as the full live engine. Factions, members, resources, goals, character facts, relationship state, threads, and memories can all overlap here. That overlap is where duplicate or drifted long-roleplay facts can appear.

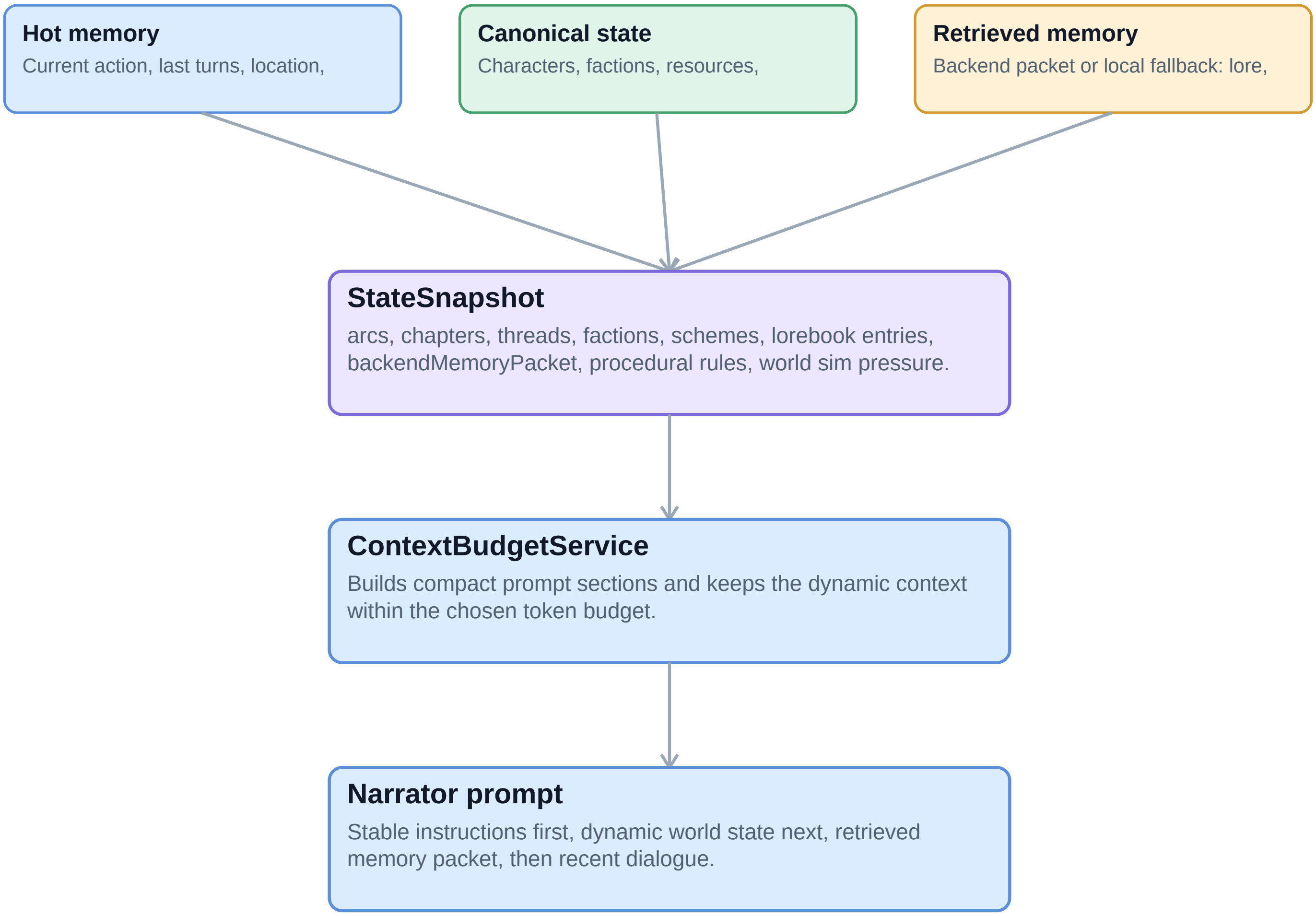
Backend Canon Data Model



Backend model note

The backend shape is cleaner than the browser model. Its strongest idea is traceability: derived canon and memory should always point back to transcript entries, state patches, or story events.

Prompt And Memory Assembly



Why this matters

- The prompt is only as reliable as the snapshot and retrieval packet feeding it.
- NPC omniscience should be stopped by passing NPC beliefs for present or referenced NPCs only.
- Faction clarity depends on keeping members, goals, resources, relations, and recent moves in the same canonical packet.
- Long-roleplay recall depends on source-linked events and memory nodes, not broad chapter prose alone.

Current Fault Lines

Turn ownership is split

The browser path still streams narration and writes local state. /api/turn exists, but currently acts more like a backend envelope

Faction data is duplicated by shape

Local lorebook faction state stores goals, resources, knownMembers, standing, and relations inside Entry.state. Backend has a

Memory is hybrid

Local retrieval still pulls lore, chapters, conversation memory, and procedural rules. Backend retrieval can build compact packets,

Transcript is evidence, not canon

The transcript should prove what happened. Canon should live in validated entities, events, patches, agreements, threads,

Long roleplay needs source-linked compression

Chapters and arcs should summarize source events while preserving traceability. Summaries that summarize summaries without

Recommended ownership model

- Dexie: offline cache, recent transcript, queued operations, temporary playability.
- Backend: canonical entities, factions, resources, memberships, beliefs, events, patches, memory nodes, sync versioning.
- Prompt builder: consume backend-built memory first, then local fallback only when offline.